Predicting Depressive Symptoms Among University Students Using Interpretable Machine Learning Models

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Section: B

1) Statement of Purpose:

Mental health is one of the most important factors of academic success that we have seen, yet the psychological well-being of university students remains one of the most under-addressed challenges in modern education. The university years are a critical development phase, but for many students this period is hampered by depressive symptoms that degrade the academic performance, motivation, etc. Old approaches to identifying depression, such as self-reported questionnaires or surveys are largely reactive and often fail to detect stress related problems at an early stage. Stigma, ego, limited access to campus mental-health resources or programs, and delays in seeking professional help further widen the gap between symptoms taking effect and meaningful intervention.

Recently it has been seen that Artificial Intelligence presents an opportunity to move from reactive identification to proactive screening. By learning patterns from demographice, academic, psychological, and behavioral data, models can uncover early indicators of depression that are not seen through traditional observations. However, many ML models are "black boxes," which limits their use in sensitive domains such as mental health. This research therefore emphasizes '**interpretable machine learning**', employing explainability methods to make sure that predictions are transparent and meaningful. Such an approach enables counselors and educators understand not only *who* is at risk, but *why*. The more accurate the WHY is, the better it is.

During my university experience, I have observed capable peers struggle silently under academic pressure, social isolation, and family issues, often without seeking help until their academic standing or personal well-being had already declined. One of my friends himself faced academic problems due to his depressive episodes which became more and more evident throughout the years. But he never talked about it until later. These experiences made me aware that reliance on self-reporting alone is insufficient, and early identification could prevent serious academic setbacks and distress.

From an academic point of view, my interest for this research came because I like applying machine learning to socially impactful problems. In summary, this study aims to contribute toward a more supportive, responsive academic environment, by allowing earlier identification of depressive symptoms in a transparent manner. Here technology complements humans rather than replacing it.

2) Research Proposal:

i. Introduction:

Mental health has become such an important concern in higher education, particularly among university students who deal with academic pressure, social transition, financial uncertainty, and future career anxiety or family issues. Nevertheless, student depression frequently goes without any treatment, often due to societal stigma, insufficient awareness, and restricted availability of profesional mental health resources.

Recent accomplishments seen in Computer Science, specifically in Artificial Intelligence (AI) and Machine Learning (ML), present interesting new methods for solving this issue. Machine learning models have the ability to analyze complex and large datasets, thus identifying patterns and risk indicators that might avoid detection through traditional methodologies. When implemented carefully, ML holds promise for enabling early screening and risk assessment for mental health disorders, including depression.

However, a key challenge in applying ML to mental health is the lack of clarity of many predictive models. Black-box predictions are difficult to trust in sensitive domains like mental health, where understanding the reasoning behind a decision is important for ethical, clinical, and practical approval. This research therefore focuses on understandable machine learning, aiming to balance predictive performance with transparency.

By creating and evaluating interpretable ML models for predicting depressive symptoms among university students, this research seeks to contribute to the field of Computer Science by showing how data-driven, explainable systems can be applied to socially significant problems. The study aims to bridge the gap between technical innovation and real-world applicability, highlighting the role of Computer Science in supporting mental health.

ii. Literature Review

Depression is widely seen as one of the worst mental health disorder targeting emotional, cognitive, behavioral, and somatic symptoms rather than a mood disturbance (Paykel, 2008). Clinical literature emphasizes that depressive disorders exist on a spectrum, ranging from subclinical forms to severe episodes associated with moments where the person can’t function and suicidal ideation (Gautam et al., 2017; Mofatteh, 2021). In South Asia, particularly India and neighboring areas, depression among young adults has been reported with increasing affect, compounded by sociocultural stigma, limited mental health literacy, and delayed help-seeking behavior (Luthra, 2017). These foundational studies establish depression as a complex condition requiring multidimensional assessment frameworks that extend beyond normal survey checklists.

University students constitute a high-risk population due to academic pressure. Empirical studies consistently demonstrate that depression negatively affects academic productivity, concentration, and overall educational outcomes (Hysenbegasi et al., 2005). Recent evidence further highlights the heterogeneity of depressive symptom profiles within student populations. Using latent profile analysis, Dong et al. (2024) identified distinct subgroups of depressive severity among Chinese university freshmen.

Social suport has been identified as a critical feature, with classmates support often making a stronger buffering effect than family support during the university years (Alsubaie et al., 2019). However, family environment remains a foundational decisive factor; adverse family environments have been directly linked to higher depressive symptoms among undergraduates in India (Panda, 2016). Gender differences are also a factor! with evidence indicating that loneliness and social isolation exert differential effects on depressive symptoms across genders, particularly intensifying symptoms among female students (Liu H. et al., 2020). Economic factors further compound vulnerability, as students from lower socioeconomic backgrounds report reduced life satisfaction and elevated depressive symptoms (Jun & Jung, 2022; Dixon & Kurpius, 2008).

To answer these challenges, recent literature has more n more adopted machine learning (ML) approaches to predict depressive symptoms among university students. Traditinal ML algorithms, including logistic regression, support vector machines (SVM), random forests, and ensemble methods, have been widely applied for depression classification and risk prediction (Ding et al., 2020; Nayan et al., 2022). While these studies rely on manually engineered features from survey data, they remain effective for table-like datasets with limited sample sizes. Comparison studies conducted in Bangladesh indicate that ensmble-based models often outperform linear classifiers in terms of accuracy and sensitivity, when predicting moderate to severe depressive symptoms (Nayan et al., 2022).

Modern machine learning methods do more than just use survey answers. They also study how people use their smartphones, how they move, and their daily behavior. By using this information, these systems can sometimes predict depression several weeks before it happens (Chikersal et al., 2021). These models use robust feature selection techniques to improve generalization and reduce noise, marking a move from reactive diagnosis toward proactive mental health monitoring. Similarly, deep and hybrid ML architectures, including integreted SVM frameworks, have shown better performance by capturing complx, non-linear relationship between predictors (Ding et al., 2020).

Even with this progress, many challenges still remain. These models often do not work equally well for everyone because the data usually comes from specific places or groups and may be unbalanced (Liu X. et al., 2022). There are also serious concerns about protecting personal data, explaining how the models make decisions, and safely using these tools in mental-health settings.

Overall, the literature demonstrates a clear movement from descriptive analyses of student depression toward predictive, machine-learning-based frameworks. While ML models show strong evidence for early screening and risk prediction, their effectiveness is dependent on data quality, interpretability, and ethical deployment. Future reserch must therefore focus on integrating psychosocial theory with robust, explainable ML models to support scalable and responsible mental-health screening among university students.

iii. Objectives

Main Objective

The main objective of this research is to develop and evaluat interpretable machine learning models for predicting depresive symptoms among university students, using demographic, academic, psychological, and lifestyle-related data.

Sub-objectives

1. To identify the most significant factors associated with depressive symptoms among university students.
2. To build and compare multiple supervised machine learning models for depression prediction.
3. To evaluate the predictive performance of these models using appropriate classification metrics.
4. To apply interpretability techniques (such as SHAP or similar methods) to explain model predictions.

iv. Research Questions

Main Research Question

How can interpretable machine learning models be used to predict depressive symptoms among university students in a transparent and reliable manner?

Sub-questions

1. What demographic, academic, psychological, and lifestyle factors are most strongly associated with depressive symptoms among university students?
2. How do different machine learning algorithms compare in terms of predictive accuracy and reliability for depression prediction?
3. Which features contribute most significantly to model predictions, and how can these contributions be interpreted meaningfully?

4. Can interpretable machine learning models provide insights that are suitable for supporting early mental health screening in university environments?

3) Proposed Research Methodology:

Research Design

This study uses a data-driven approach based on supervised machine learning. Data will be collected through surveys, then cleaned and prepared for analysis. After that, important factors will be examined, prediction models will be built and tested, and the results will be interpreted. The main focus of the study is to build accurate, transparent, and responsible models related to student mental health.

Research Question 1: What demographic, academic, psychological, and lifestyle factors are most strongly associated with depressive symptoms among university students?

Methodology:
For this question, the study will explore the data and pick out important features. Survey responses from university students will inclede information such as age and gender, academic stress, anxiety and stress levels, sleep quality, and physical activity.

First, simple statistical tests and correlation analysis will be used to see how each factor relates to depression scores. Then, machine learning models like Logistic Regression and Random Forest will be applied to identify which factors matter the most. Model outputs such as coefficients and feature importance scores will explain the influence of each factor.

This method is used because it combines basic statistical analysis with machine learning, making it easier to find both simple relationships and more complex patterns in the data.

Research Question 2: How do different machine learning algorithms compare in terms of predictive accuracy and reliability for depression prediction?

Methodology:
To answer this question, several supervised machine learning models will be trained and compared. These include Logistic Regression, Support Vector Machine (SVM), Random Forest, and Gradient Boosting. The dataset will be split into training and testing parts, and cross-validation will be used to make the results more reliable.

The models will be evaluated using accuracy, precision, recall, F1-score, and AUC. More importance will be given to recall, since missing students who may be experiencing depression can have serious consequences in mental health screening.

Research Question 3: Which features contribute most significantly to model predictions, and how can these contributions be interpreted meaningfully?

Methodology:
Here interpretability techniques will be applied to the best-performing models. SHAP (SHapley Additive exPlanations) will be used to show how each feature affects the model's predictions.

The analysis will include overall explanations to show which factors are most important across all students, as well as individual explanations to show why a specific student was predicted to be at risk or not. This makes the model easier to understand and trust.

This step is important because mental health-related predictions need to be transparent and explainable.

Research Question 4: Can interpretable machine learning models provide insights that are suitable for supporting early mental health screening in university environments?

Methodology:
The study will focus on building models that are easy to explain, such as Logistic Regression or other interpretable models, while still maintaining good prediction performance.

The final model will be evaluated not only based on accuracy and other metrics, but also on how easy it is to understand and use in real situations. The explanations produced by the model will be checked to see if they make sense and match existing mental health research.

This approach is chosen because early screening tools must be both reliable and easy for educators or health professionals to understand and apply responsibly.

4) Ethical Considerations:

Mental-health research topics are very sensitive. Particularly research involving depressive symptoms and suicidal tendency among university students, strong ethical safeguards are essential. This directly affects a human-being and we have to keep this in mind. This study emphasizes on data protection, confidentiality, and anonymity to ensure that participants' rights and well-being are fully respected throughout the research lifecycle.

Data protection will be ensured through strict data governance practices. All dataasets used in this research whether collected directly through surveys or obtained from secondary sources will be stored in encrypted, controlled environments. Only the principl researcher will have access to raw data, and no data will be stored on unsecured personal devices. These practices align with widely accepted ethical guidelines for handling sensitive psychological and health-related data (Chikersal et al., 2021; Liu X. et al., 2022).

Confidentiality will be kept by ensuring that no personally identifiable information (PII), such as names, student IDs, email addresses, or institutional names, is collected or kept. Prior studies in student depression prediction emphasize that maintaining confidentiality is critical for participant trust and data reliability (Ding et al., 2020; Nayan et al., 2022).

Anonymity will be ensured by assigning random, non-traceable identifiers to all data records before analysis. Once preprocessing is complete, original datasets will be removed of any indirect identifiers that's risky. This approach reduces the risk of psychological harm, which is particularly important in cultures where mental-health stigma remains prevalent (Luthra, 2017; Gautam et al., 2017).

Overall, by prioritizing secure data handling, strict confidentiality, and full anonymity, this research aims to uphold ethical integrity while contributing to the application of machine learning in mental-health screening among university students.

5) References:

- Alsubaie, M. M., Stain, H. J., Webster, L. A. D., & Sjaugi, M. F. (2019). The role of sources of social support on depression and quality of life for university students. International Journal of Adolescence and Youth, 24(4), 484–496. https://doi.org/10.1080/02673843.2019.1568887
- Chikersal, P., Doryab, A., Tumminia, M., et al. (2021). Detecting depression and predicting its onset using longitudinal symptoms captured by passive sensing: A machine learning approach with robust feature selection. ACM Transactions on Computer-Human Interaction, 28(1), 1–41. https://doi.org/10.1145/3422821

- Ding, Y., Chen, X., Fu, Q., & Zhong, S. (2020). A depression recognition method for college students using deep integrated support vector algorithm. IEEE Access, 8, 75616–75629. https://doi.org/10.1109/ACCESS.2020.2987523
- Dixon, S. K., & Kurpius, S. E. R. (2008). Depression and college stress among university undergraduates: Do mattering and self-esteem make a difference? Journal of College Student Development, 49(5), 412–424. https://doi.org/10.1353/csd.0.0024
- Dong, J., Li, Y., Chen, J., et al. (2024). The relationship between depression heterogeneity and academic performance in Chinese university freshmen: A latent profile analysis. Current Psychology, 1–12. https://doi.org/10.1007/s12144-024-05795-y
- Fang, J., Wang, T., Li, C., Hu, X., Ngai, E., Seet, B., et al. (2019). Depression prevalence in postgraduate students and its association with gait abnormality. IEEE Access, 7, 174425–174437. https://doi.org/10.1109/ACCESS.2019.2957179
- Gautam, S., Jain, A. K., Gautam, M., Vahia, V., & Grover, S. (2017). Clinical practice guidelines for the management of depression. Indian Journal of Psychiatry, 59, S34–S50. https://doi.org/10.4103/0019-5545.203743
- Hysenbegasi, A., Hass, S. L., & Rowland, C. R. (2005). The impact of depression on the academic productivity of university students. The Journal of Mental Health Policy and Economics, 8(3), 145–151.
- Jun, S. K., & Jung, I. S. (2022). Exploration of the differences in positive-negative psychological factors influencing the life satisfaction and depression of college students by gender and family economic level. International Journal of Advanced and Applied Sciences, 9(4), 88–96. https://doi.org/10.21833/ijaas.2022.04.011
- Liu, H., Zhang, M., Yang, Q., & Yu, B. (2020). Gender differences in the influence of social isolation and loneliness on depressive symptoms in college students: A longitudinal study. Social Psychiatry and Psychiatric Epidemiology, 55, 251–257. https://doi.org/10.1007/s00127-019-01726-6
- Liu, X., Guo, Y., Zhang, W. J., & Gao, W. (2022). Influencing factors, prediction and prevention of depression in college students: A literature review. World Journal of Psychiatry, 12(7), 860–873. https://doi.org/10.5498/wjp.v12.i7.860
- Liu, Y., et al. (2025). Predictors of depression among Chinese college students: A machine learning approach. BMC Public Health, 25(1). https://doi.org/10.1186/s12889-025-21632-8
- Luthra, M. (2017). Depression: Current scenario with reference to India. Indian Journal of Community Health, 29(1), 1–4. https://doi.org/10.47203/IJCH.2017.v29i01.001
- Margitics, F. (2005). Predisposing factors and subclinical depression among healthy college students. Psychiatria Hungarica, 20(3), 211–223.
- Mofatteh, M. (2021). Depressive and other symptoms of emotional disorders in adolescents. International Journal of Social Psychiatry, 57(S1), 22–39. https://doi.org/10.1177/0020764010396695
- Moreno-Küstner, B., Jones, R., Svab, I., Maaroos, H., Xavier, M., Geerlings, M., et al. (2016). Suicidality in primary care patients who present with sadness and anhedonia: A prospective European study. BMC Psychiatry, 16(1), 1–10. https://doi.org/10.1186/s12888-016-0786-5
- Nayan, M., Uddin, M., Hossain, M., Alam, M., Zinnia, M., Haq, I., et al. (2022). Comparison of the performance of machine learning-based algorithms for predicting depression and anxiety among university students in Bangladesh. Asian Journal of Social Health and Behavior, 5(2), 75–84. https://doi.org/10.4103/shb.shb_38_22
- Panda, S. (2016). Depression among undergraduate college students in relation to their family environment. Indian Journal of Health and Wellbeing, 7(4), 478–482.
- Park, U. S., & Cho, Y. C. (2018). Depression symptom levels and its related factors among college students. Journal of the Korea Academia-Industrial Cooperation Society, 19(6), 219–230. https://doi.org/10.5762/KAIS.2018.19.6.219
- Paykel, E. S. (2008). Basic concepts of depression. Dialogues in Clinical Neuroscience, 10(3), 279–289. https://doi.org/10.31887/DCNS.2008.10.3/espaykel